

## 4.10

**Theorem 4.19** Let  $f$  be a continuous mapping of a compact metric space  $X$  into a metric space  $Y$ . Then  $f$  is uniformly continuous on  $X$ .

*Proof.* Assume, for contradiction, that  $f$  is not uniformly continuous. Then, there exists  $\epsilon > 0$  such that for all  $\delta > 0$  there exists  $p \in X$  such that  $f(B_\delta(p, X)) \not\subset B_\epsilon(f(p), Y)$ . Let  $q \in B_\delta(p, X)$  such that  $f(q) \in f(B_\delta(p, X)) \setminus B_\epsilon(f(p), Y)$ . So, we have  $d_X(p, q) < \delta$ , and  $d_Y(f(p), f(q)) > \epsilon$ . Construct sequences  $(p_n)$  and  $(q_n)$  corresponding to the sequence  $\delta_n = \frac{1}{n}$  such that  $d_X(p_n, q_n) < \frac{1}{n}$ . Thus,  $d_X(p_n, q_n) \rightarrow 0$ ,  $d_Y(f(p_n), f(q_n)) > \epsilon$ .

From Theorem 2.37,  $\{p_n\}$  has a limit point  $l \in X$ . Thus, there exists a subsequence  $(p_{n_k})$  of  $(p_n)$  which converges to  $l$ . Because  $d_X(p_n, q_n) \rightarrow 0$ , it follows that  $(q_{n_k}) \rightarrow l$ . By the continuity of  $f$  over  $X$ ,  $\lim_{x \rightarrow l} f(x) = f(l)$ . The sequence criterion tells us that  $(p_{n_k}) \rightarrow l$  and  $(q_{n_k}) \rightarrow l$  imply that  $(f(p_{n_k})) \rightarrow f(l)$  and  $(f(q_{n_k})) \rightarrow f(l)$ . However, this is impossible since  $d_Y(f(p_{n_k}), f(q_{n_k})) > \epsilon$ . This contradiction forces us to conclude  $f$  is uniformly continuous.  $\square$

## 4.13

$E$  is a dense subset of  $X$ .

$f : E \rightarrow \mathbb{R}$  is uniformly continuous.

**To prove:**  $f$  has a continuous extension from  $E$  to  $X$ .

*Proof.* Because  $E$  is dense in  $X$ , every point of  $X$  is either in  $E$  or a limit point of  $E$ . So, all points in  $X \setminus E$  are limit points of  $E$ . Consider  $p \in X \setminus E$ . There exists a sequence  $(p_n) \rightarrow p$  in  $E$ . Let  $\epsilon > 0$ . Since  $f$  is uniformly continuous, there exists  $\delta$  such that  $d_X(p_n, p_m) < \delta$  implies  $|f(p_n) - f(p_m)| < \epsilon$ . Since convergent sequences are Cauchy sequences in metric spaces, there exists  $N$  such that  $n, m > N$  implies  $d_X(p_n, p_m) < \delta$ . Thus,  $n, m > N$  implies  $|f(p_n) - f(p_m)| < \epsilon$ , i.e.,  $(f(p_n))$  is Cauchy. Since  $\mathbb{R}$  is complete,  $(f(p_n))$  must converge to some  $l_p \in \mathbb{R}$ .

Consider another sequence  $(q_n) \rightarrow p$ . It follows that  $d_X(p_n, q_n) \rightarrow 0$ . Following the same chain of reasoning as in the previous paragraph leads us to conclude  $|f(p_n) - f(q_n)| \rightarrow 0$ . Thus,  $f(q_n) \rightarrow l_p$ . We have shown that for every sequence  $(q_n) \rightarrow p$ ,  $f(q_n) \rightarrow l_p$ . Thus,  $\lim_{x \rightarrow p} f(x) = l_p$ . Now, define  $F : X \rightarrow \mathbb{R}$  by

$$F(x) = \begin{cases} f(x) & x \in E \\ l_x & x \notin E \end{cases}$$

We will now show that  $F$ , as defined, is continuous. Let  $\epsilon > 0$ . Let  $s \in X$ . If  $s \in E$ , since  $f$  is continuous at  $s$ , there exists  $\delta > 0$  such that  $f(B_\delta(s, E)) \subset B_{\frac{\epsilon}{2}}(f(s), \mathbb{R})$ . If  $s \in X \setminus E$ ,  $\lim_{x \rightarrow s} f(x) = F(s)$  by definition, so there exists  $\delta > 0$  such that  $f(B_\delta(s, E) \setminus \{s\}) \subset B_{\frac{\epsilon}{2}}(F(s), \mathbb{R})$ . In either case, it follows from our definition of  $F$  that  $F(B_\delta(s, X)) \subset \overline{B_{\frac{\epsilon}{2}}(F(s), \mathbb{R})} \subset B_\epsilon(F(s), \mathbb{R})$ . Thus,  $F$  is continuous at  $s$ .  $\square$

## 4.15

$f : X \rightarrow Y$  is *open* if for all open  $V \subset X$ ,  $f(V)$  is open in  $Y$ .

**To prove:** Every continuous open mapping of  $\mathbb{R}$  into  $\mathbb{R}$  is monotonic.

*Proof.* Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be continuous and open. FTSOC, assume  $f$  is not monotonic. Then, there exist  $x_1 < x_2 < x_3$  such that  $f(x_1) < f(x_2) > f(x_3)$  (the other case  $f(x_1) > f(x_2) < f(x_3)$  follows similarly). Since  $[x_1, x_3]$  is compact,  $f(x)$  attains a maximum value in  $[x_1, x_3]$ , say  $m$ , courtesy the extreme value theorem. We know that  $m \neq f(x_1)$  and  $m \neq f(x_3)$  since  $m \geq f(x_2)$ . Thus,  $f(x)$  attains the maximum value  $m$  in  $(x_1, x_3)$ . Consider the image  $I = f((x_1, x_3))$ . We know that  $m = \sup I$  and  $m \in I$ . This makes it impossible for  $I$  to contain a neighborhood of  $m$ . Thus,  $I$  is not open, which contradicts our hypothesis that  $f$  is an open map. Therefore,  $f$  must be monotonic.  $\square$

## 2.19

- (a) **To prove:** If  $A$  and  $B$  are disjoint closed sets in some metric space  $X$ , they are separated.

*Proof.* Since  $A$  and  $B$  are closed in  $X$ ,  $\overline{A} = A$  and  $\overline{B} = B$ . We are given that  $A \cap B = \emptyset$ . It follows that  $\overline{A} \cap B = \emptyset$  and  $A \cap \overline{B} = \emptyset$ .  $\square$

- (b) **To prove:** If  $A$  and  $B$  are disjoint open sets in some metric space  $X$ , they are separated.

*Proof.* FTSOC, assume  $p \in \overline{A} \cap B$ . Since  $A \cap B = \emptyset$ ,  $p \notin A$ . So,  $p$  is a limit point of  $A$ . It follows that for all  $\epsilon > 0$ ,  $B_\epsilon(p, X) \cap A \neq \emptyset$ , i.e.,  $B$  does not contain a neighborhood of  $p$ . This contradicts our hypothesis that  $B$  is open. Therefore, we must have  $\overline{A} \cap B = \emptyset$ . Similarly, we can show that  $A \cap \overline{B} = \emptyset$ .  $\square$

- (c) **To prove:** For  $p \in X$ ,  $\delta > 0$ , define  $A = \{q \mid d(p, q) < \delta\}$ ,  $B = \{q \mid d(p, q) > \delta\}$ .  $A$  and  $B$  are separated.

*Proof.*  $A$  is an open ball. Take any point  $q \in B$ . Let  $\gamma \equiv d(q, p)$  and  $r \equiv \frac{\gamma - \delta}{2}$ . Then,  $B_r(q, X) \subset B$ . Thus,  $B$  is also open. Clearly,  $A$  and  $B$  are disjoint. It follows from (b) that  $A$  and  $B$  are separated.  $\square$

- (d) **To prove:** Every connected metric space with at least two points is uncountable.

*Proof.* Let  $X$  be a connected metric space. Let  $x, y \in X$ ,  $x \neq y$ . Let  $r \equiv d(x, y)$ . Let  $\alpha \in (0, r)$ . Define  $A \equiv \{q \mid d(q, x) < \alpha\}$  and  $B \equiv \{q \mid d(q, x) > \alpha\}$ . Since  $A \sqcup B$  cannot be a separation of  $X$ , there must exist  $z \in X$  such that  $d(x, z) = \alpha$ . Thus, with every  $\alpha \in (0, r)$  there must be associated a unique  $z_\alpha \in X$ . We know that  $(0, r)$  is uncountable. It follows that  $X$  is uncountable.  $\square$

## 5.1

$$\begin{aligned} f &: \mathbb{R} \rightarrow \mathbb{R}, \\ |f(x) - f(y)| &\leq (x - y)^2 \quad \forall x, y \in \mathbb{R}. \end{aligned}$$

**To prove:**  $f$  is constant.

*Proof.* Let  $\Delta x \equiv y - x$ . Then,  $y = x + \Delta x$ . This gives us

$$\begin{aligned} &|f(x + \Delta x) - f(x)| \leq \Delta x^2 \\ \implies &\frac{|f(x + \Delta x) - f(x)|}{|\Delta x|} \leq |\Delta x| \\ \implies &\lim_{\Delta x \rightarrow 0} \left| \frac{f(x + \Delta x) - f(x)}{\Delta x} \right| \leq \lim_{\Delta x \rightarrow 0} |\Delta x| \\ \implies &|f'(x)| \leq 0 \\ \implies &f'(x) = 0. \end{aligned}$$

Thus,  $f'(x) = 0$  for all  $x \in \mathbb{R}$ . FTSOC, assume  $f$  is not constant, i.e, there exist  $x_1$  and  $x_2$  such that  $f(x_1) \neq f(x_2)$ . WLOG,  $x_1 < x_2$ . Then, from the mean value theorem, there exists  $c$  such that  $x_1 < c < x_2$  and  $f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} \neq 0$ . However,  $f'(x) = 0$  for all  $x$ . This contradiction implies that  $f$  is indeed a constant function.  $\square$